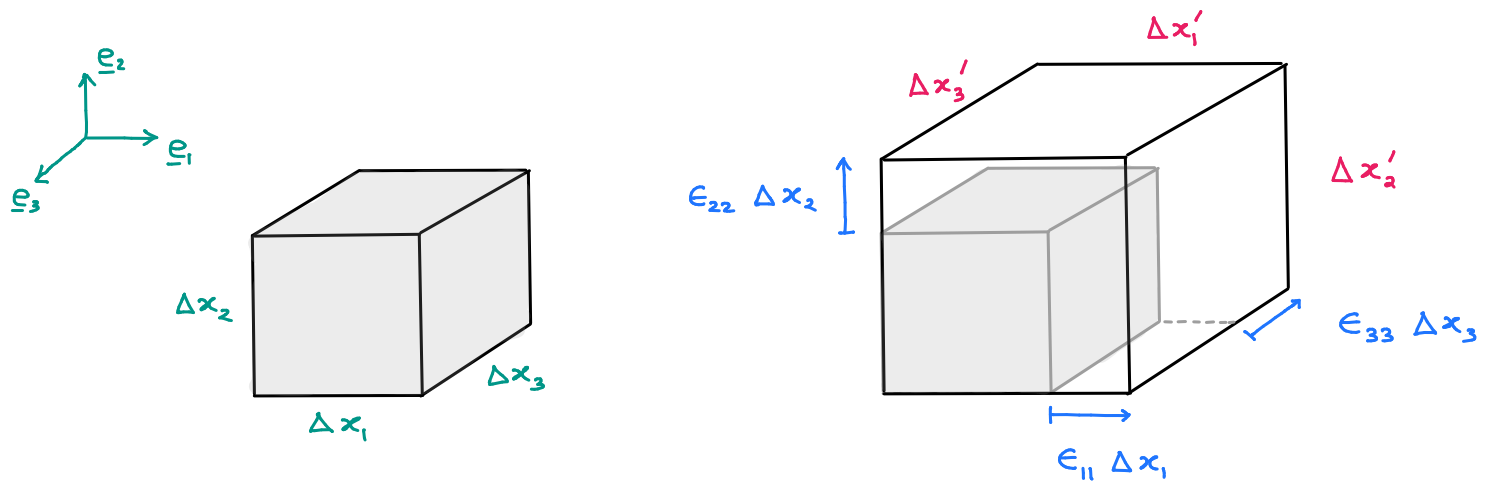


## Local Volumetric Strain (or Dilatational Strain)

As a body deforms, the volume of every small region (called local volume element) of the body also changes. We can then define a quantity called volumetric strain because the change in volume per unit volume will be different for different parts in a body.

Normal strain leads to change in volume, so we will consider a small local volume element (in the form of a cuboid) at a point in the undeformed body.



$$\left. \begin{aligned} \Delta x'_1 &= \Delta x_1 + \epsilon_{11} \Delta x_1 \\ \Delta x'_2 &= \Delta x_2 + \epsilon_{22} \Delta x_2 \\ \Delta x'_3 &= \Delta x_3 + \epsilon_{33} \Delta x_3 \end{aligned} \right\} \text{ for small strains}$$

Volume of original local volume element,  $V = \Delta x_1 \Delta x_2 \Delta x_3$

Volume of the element after deformation,  $V' = \Delta x'_1 \Delta x'_2 \Delta x'_3$

$$\begin{aligned} \text{Volumetric strain} &= \frac{V' - V}{V} = \frac{(1 + \epsilon_{11})(1 + \epsilon_{22})(1 + \epsilon_{33}) V - V}{V} \\ &\approx \epsilon_{11} + \epsilon_{22} + \epsilon_{33} \quad (\text{neglecting products of small strains}) \end{aligned}$$

## Local average rotation tensor

Note that the displacement gradient  $\underline{\nabla} \underline{u}$  can be written as:

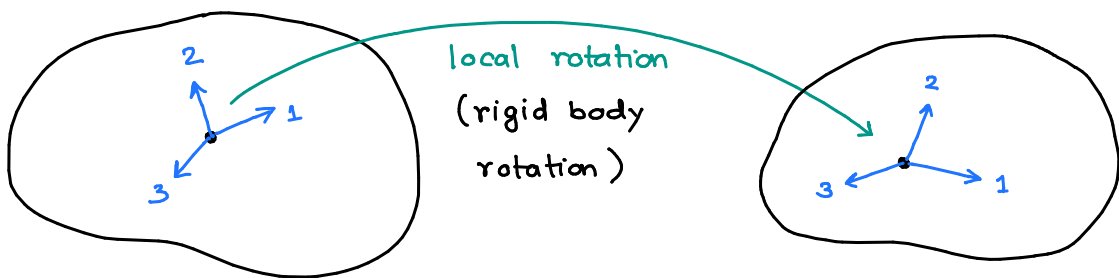
$$\underline{\nabla} \underline{u} = \frac{1}{2} (\underline{\nabla} \underline{u} + \underline{\nabla} \underline{u}^T) + \frac{1}{2} (\underline{\nabla} \underline{u} - \underline{\nabla} \underline{u}^T)$$

$$= \underline{\underline{\epsilon}} + \underline{\underline{W}}$$

$\underline{\underline{\epsilon}}$   
small strain tensor  
(induces strain at a point)

$\underline{\underline{W}}$   
local average rotation tensor

(causes rigid-body rotation of line elements at a pt)



ANTI-SYMMETRIC MATRIX

$$[\underline{\underline{W}}]_{\begin{pmatrix} e_1 \\ e_2 \\ e_3 \end{pmatrix}} = \begin{bmatrix} 0 & \frac{1}{2} \left( \frac{\partial u_1}{\partial x_2} - \frac{\partial u_2}{\partial x_1} \right) & \frac{1}{2} \left( \frac{\partial u_1}{\partial x_3} - \frac{\partial u_3}{\partial x_1} \right) \\ -\frac{1}{2} \left( \frac{\partial u_1}{\partial x_2} - \frac{\partial u_2}{\partial x_1} \right) & 0 & \frac{1}{2} \left( \frac{\partial u_2}{\partial x_3} - \frac{\partial u_3}{\partial x_2} \right) \\ -\frac{1}{2} \left( \frac{\partial u_1}{\partial x_3} - \frac{\partial u_3}{\partial x_1} \right) & -\frac{1}{2} \left( \frac{\partial u_2}{\partial x_3} - \frac{\partial u_3}{\partial x_2} \right) & 0 \end{bmatrix}$$

The local average rotation tensor is responsible for rigid-body rotation of line elements. So if the displacement is such that the strain tensor  $\underline{\underline{\epsilon}}$  is  $\underline{\underline{0}}$  at a point, then there will be no strain of any kind (normal/shear strain) at that point. However, due to  $\underline{\underline{W}}$ , the line elements may undergo rigid rotation.  $\underline{\underline{W}}$  can vary from point to point, meaning rigid body rotation will be different for different points in the body.

## Strain-compatibility equations

We have seen that displacement of a point in a deformable body is a vector  $\underline{u}$  which has three components.

$$[\underline{u}] \begin{pmatrix} e_1 \\ e_2 \\ e_3 \end{pmatrix} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}$$

The deformation at a point is specified by six independent strain components

$$\epsilon_{11}, \epsilon_{22}, \epsilon_{33}, \epsilon_{12}, \epsilon_{13}, \epsilon_{23}$$

We have seen from strain-displacement relations that the six strains can be determined from three displacement functions, since they only involve differentiation of the displacement. Example:

$$\epsilon_{11} = \frac{\partial u_1}{\partial x_1}, \quad \epsilon_{22} = \frac{\partial u_2}{\partial x_2}, \quad \epsilon_{12} = \frac{1}{2} \left( \frac{\partial u_1}{\partial x_2} + \frac{\partial u_2}{\partial x_1} \right)$$

However, the reverse operation, i.e. determination of the three displacement functions  $u_1, u_2, u_3$  from six strain components is not straightforward. Infact, we may not obtain a consistent displacement function by integrating any six arbitrary strain components

For example, think of integrating  $\epsilon_{11}(x_1, x_2, x_3)$  w.r.t  $x_1$  to obtain  $u_1$

$$\Rightarrow u_1(x_1, x_2, x_3) = \int \epsilon_{11}(x_1, x_2, x_3) dx_1 + c_1(x_2, x_3)$$

Then integrating  $\epsilon_{22}(x_1, x_2, x_3)$  w.r.t  $x_2$  should give  $u_2$

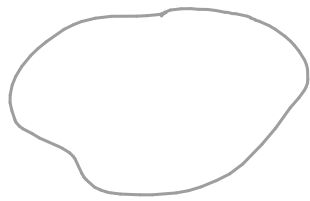
$$\Rightarrow u_2(x_1, x_2, x_3) = \int \epsilon_{22}(x_1, x_2, x_3) dx_2 + c_2(x_1, x_3)$$

The resulting functions  $u_1$  and  $u_2$  obtained from integration of  $\epsilon_{11}$  and  $\epsilon_{22}$  maynot satisfy the prescribed function for  $\epsilon_{12}$

Since 6 strain components are derived from 3 displacements, the six strain components cannot be arbitrary, else six arbitrary strain components could lead to "incompatible" displacement.

Physically, it may happen that the displacement so obtained is such that it leads to two parts of a body overlapping with each other or getting separated

Undeformed body



Incompatible  
displacements  
of the body

Deformed body



Thus, there has to be some constraint on the six strain functions which are collectively called **STRAIN COMPATIBILITY** conditions

Remember that a general symmetric matrix does not necessarily represent a strain matrix until it satisfies strain compatibility conditions.

There are **SIX** strain-compatibility equations (much like three stress-equilibrium relations), which can be divided into two sets:

$$\boxed{\text{Set 1}} \left\{ \begin{array}{l} \frac{\partial^2 \epsilon_{11}}{\partial x_2^2} + \frac{\partial^2 \epsilon_{22}}{\partial x_1^2} = 2 \frac{\partial^2 \epsilon_{12}}{\partial x_1 \partial x_2} \\ \frac{\partial^2 \epsilon_{22}}{\partial x_3^2} + \frac{\partial^2 \epsilon_{33}}{\partial x_2^2} = 2 \frac{\partial^2 \epsilon_{23}}{\partial x_2 \partial x_3} \\ \frac{\partial^2 \epsilon_{11}}{\partial x_3^2} + \frac{\partial^2 \epsilon_{33}}{\partial x_1^2} = 2 \frac{\partial^2 \epsilon_{13}}{\partial x_1 \partial x_3} \end{array} \right.$$

We can verify the relations of set 1 by plugging in strain-disp relation.

$$\begin{aligned} \text{LHS} &= \frac{\partial^2 \epsilon_{11}}{\partial x_2^2} + \frac{\partial^2 \epsilon_{22}}{\partial x_1^2} = \frac{\partial^3 u_1}{\partial x_2^2 \partial x_1} + \frac{\partial^3 u_2}{\partial x_1^2 \partial x_2} \\ &= \frac{\partial^2}{\partial x_1 \partial x_2} \left( \frac{\partial u_1}{\partial x_2} + \frac{\partial u_2}{\partial x_1} \right) = 2 \frac{\partial^2 \epsilon_{12}}{\partial x_1 \partial x_2} = \text{RHS} \end{aligned}$$

Similarly, you can prove others

The second set of compatibility conditions is:

$$\frac{\partial}{\partial x_3} \left( \frac{\partial \epsilon_{23}}{\partial x_1} + \frac{\partial \epsilon_{31}}{\partial x_2} - \frac{\partial \epsilon_{12}}{\partial x_3} \right) = \frac{\partial^2 \epsilon_{33}}{\partial x_1 \partial x_2}$$

$$\frac{\partial}{\partial x_1} \left( \frac{\partial \epsilon_{12}}{\partial x_3} + \frac{\partial \epsilon_{13}}{\partial x_2} - \frac{\partial \epsilon_{23}}{\partial x_1} \right) = \frac{\partial^2 \epsilon_{11}}{\partial x_2 \partial x_3}$$

$$\frac{\partial}{\partial x_2} \left( \frac{\partial \epsilon_{12}}{\partial x_3} + \frac{\partial \epsilon_{23}}{\partial x_1} - \frac{\partial \epsilon_{13}}{\partial x_2} \right) = \frac{\partial^2 \epsilon_{33}}{\partial x_1 \partial x_2}$$

### Special case for plane strain

For the case where strains are significant only on a plane, say  $\epsilon_1$ - $\epsilon_2$  (or  $x$ - $y$ ) plane, we can neglect strains along the  $\epsilon_3$  directions, i.e.

$$\left. \begin{aligned} \epsilon_{11} &= \epsilon_{11}(x_1, x_2) \\ \epsilon_{22} &= \epsilon_{22}(x_1, x_2) \\ \epsilon_{12} &= \epsilon_{12}(x_1, x_2) \end{aligned} \right\} \begin{array}{l} \text{functions of } x_1 \text{ and } x_2 \\ \text{only} \end{array}$$

$$\epsilon_{33} = \epsilon_{13} = \epsilon_{23} = 0$$

For this case, five compatibility conditions are automatically satisfied. Only  $\frac{\partial^2 \epsilon_{11}}{\partial x_2^2} + \frac{\partial^2 \epsilon_{22}}{\partial x_1^2} = 2 \frac{\partial^2 \epsilon_{12}}{\partial x_1 \partial x_2}$  needs checking.